Category Theory for Mathematical Physics — Exercise Sheet 9

Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$

Variant: Hints and answers

Conventions as in Lecture 9: the crossing is $c = A\,\mathrm{id} + A^{-1}U$ (Equation (9.4)), the loop is $\delta = -A^2 - A^{-2}$ (Equation (9.3)), $q = A^2$ and $[n] = [n]_q = (q^n - q^{-n})/(q - q^{-1})$, so $\delta = -[2]$. In $\mathrm{TL}_n(\delta)$, U_i is the cup-cap on strands $i, i+1$; a product XY means Y first (drawn below). The *closure* of a diagram on n strands joins each top endpoint to the bottom one below it on the right, and is evaluated by Equation (9.4) and Equation (9.3). p_n are the Jones–Wenzl projectors (Theorem 9.7). A coloured strand is labelled by its number n of strands (its colour), as in $(2, p_2)$, θ_n, d_n; only in Exercise 4(ii)–(iii) are the simple objects of $\mathrm{SU}(2)_3$ labelled by spins $j = n/2$, as in Theorem 9.9. $\varphi = (1+\sqrt{5})/2$.

Exercise 1 (Hopf link, trefoil, Jones). (i) Draw the Hopf link as the closure of c^2 and the right-handed trefoil as the closure of c^3, on two strands, each strand oriented upward along the braid (so every crossing is positive). Write c^2 and c^3 as $\alpha + \beta U$ in $\mathrm{TL}_2(\delta)$, and close them up to get $\langle\mathrm{Hopf}\rangle$ and $\langle\mathrm{trefoil}\rangle$.

(ii) With the writhe w (the number of positive crossings minus negative ones), the Jones polynomial is $V_L(t) = (-A^3)^{-w}\langle L\rangle/\langle O\rangle$ at $A = t^{-1/4}$ (Lecture 9). Compute it for the Hopf link and the trefoil.

Hint. (i) The closure of id on two strands is two loops, that of U one loop; $U^2 = \delta U$. (ii) $\langle O\rangle = \delta$, and both links here have only positive crossings.

Answer. (i) $c^2 = A^2 + (1 - A^{-4})U$, $c^3 = A^3 + (A - A^{-3} + A^{-7})U$; $\langle\mathrm{Hopf}\rangle = \delta(-A^4 - A^{-4})$, $\langle\mathrm{trefoil}\rangle = \delta(-A^5 - A^{-3} + A^{-7})$. (ii) $V_{\mathrm{Hopf}} = -t^{1/2} - t^{5/2}$, $V_{\mathrm{trefoil}} = t + t^3 - t^4$.

Exercise 2 (TL_3 and Reidemeister III). (i) Draw the five diagrams $1, U_1, U_2, U_1U_2, U_2U_1$ on three strands, and show that the product of any two of them is a multiple of one of them. Conclude $\dim \mathrm{TL}_3 = 5$, given that the five are independent (they are, on $V_1^{\otimes 3}$ for generic A).

(ii) With $c_i = A + A^{-1}U_i$, show from the TL relations alone that

$$c_1c_2c_1 = A^3 + A(U_1 + U_2) + A^{-1}(U_1U_2 + U_2U_1) = c_2c_1c_2.$$

(iii) Deduce $c_1c_2c_1^{-1} = c_2^{-1}c_1c_2$, the braid relation with two of its crossings inverted.

Hint. (i) Use $U_i^2 = \delta U_i$ and $U_iU_jU_i = U_i$ to shorten words. (ii) Collect the coefficient of U_1; $A^2 + A^{-2} + \delta = 0$.

Answer. (i) Every word in U_1, U_2 shortens to one of the five times a power of δ (e.g. $U_1U_2 \cdot U_1 = U_1$). (ii) The coefficient of U_1 is $2A + A^{-1}\delta + A^{-3} = A$. (iii) Multiply (ii) by c_2^{-1} on the left and c_1^{-1} on the right.

Exercise 3 (The projectors p_2 and p_3). Throughout, A is such that $[2], [3] \neq 0$.

(i) Show that $p_2 = 1 - U/\delta = 1 + U/[2]$ is idempotent with $Up_2 = 0$, and that its closure is $[3]$.

(ii) Build $p_3 = p_2 \otimes 1 + \frac{[2]}{[3]}(p_2 \otimes 1)U_2(p_2 \otimes 1)$ in TL_3 and show that

$$p_3 = 1 + \frac{[2]}{[3]}(U_1 + U_2) + \frac{1}{[3]}(U_1U_2 + U_2U_1).$$

(iii) Check $U_1 p_3 = 0$, and compute the closure of p_3.

Hint. (ii) $p_2 \otimes 1 = 1 + U_1/[2]$, and $[2]^2 = [3] + 1$. (iii) The closures of 1, U_i and U_1U_2, U_2U_1 are δ^3, δ^2, δ; and $[4] = [2]^3 - 2[2]$.

Answer. (i) $p_2^2 = 1 + (2/[2] + \delta/[2]^2)U = p_2$; closure $\delta^2 - 1 = [3]$. (ii) The coefficient of U_1 is $\frac{1}{[2]} + \frac{1}{[2][3]} = \frac{[2]}{[3]}$. (iii) $U_1 p_3 = \big(1 + \delta\frac{[2]}{[3]} + \frac{1}{[3]}\big)U_1 + \big(\frac{[2]}{[3]} + \frac{\delta}{[3]}\big)U_1U_2 = 0$; the closure is $-[4]$.

Exercise 4 (Coloured twists, and Fibonacci inside $\mathrm{SU}(2)_3$). (i) Draw the twist axiom of Definition 9.1 for $V_1 \otimes V_1$ (Equation (9.1)) with p_2 inserted at the bottom, and use it, with $\theta_1 = -A^3$, to show that the coloured strand $(2, p_2)$ has twist A^8, as Equation (9.10) says.

(ii) At $k = 3$, $A = e^{i\pi/10}$: list d_j and θ_j for $j = 0, \frac{1}{2}, 1, \frac{3}{2}$ after the pivotal flip of Theorem 9.9. Show that $\{0, 1\}$ is Fibonacci, with its d_τ and θ_τ, and identify $\frac{3}{2}$.

(iii) Check that $\theta_{1/2} = \theta_1\theta_{3/2}$.

Hint. (i) $c = A + A^{-1}U$ and $Up_2 = 0$. (ii) $[3] = 1 + 2\cos(2\pi/5)$; $\frac{3}{2} \otimes \frac{3}{2} \cong 0$.

Answer. (i) $c\,p_2 = Ap_2$, so $\theta = c^2(\theta_1 \otimes \theta_1) = A^2A^6 = A^8$ on p_2. (ii) $d_j = 1, \varphi, \varphi, 1$; $\theta_j = 1, e^{3\pi i/10}, e^{4\pi i/5}, -i$: $\{0, 1\}$ is Fibonacci ($1 \otimes 1 \cong 0 \oplus 1$), and $\frac{3}{2}$ is an antisemion. (iii) $e^{4\pi i/5} \cdot e^{-i\pi/2} = e^{3\pi i/10}$.

Exercise 5 (The $k = 1$ sign check). Take $k = 1$, $A = e^{i\pi/6}$.

(i) Compute δ and $[3]$, and show that p_2 is negligible (Definition 9.8); which coloured strands survive?

(ii) Compute d_1 and θ_1 for the colour 1 (spin $\frac{1}{2}$) in the Temperley–Lieb pivotal structure and after the flip.

(iii) Compute the number by which c acts on the cup $\mathbb{C} \to V_1 \otimes V_1$, and compare with the semion of Lecture 7.

Hint. (i) $\mathrm{tr}(gp_2) = \mathrm{tr}(p_2gp_2)$, and p_2gp_2 is a multiple of p_2 (Theorem 9.7). (iii) U acts on the cup by δ.

Answer. (i) $\delta = -1$, $[3] = 0$: the closure of p_2 vanishes; 0 and 1 survive. (ii) $d_1 = -1$, $\theta_1 = -i$ (TL); $d_1 = 1$, $\theta_1 = i$ (flipped). (iii) $A + A^{-1}\delta = i$: the semion's $c_{s,s} = i = \theta_s$, in the flipped structure.

Exercise 6 ⋆. *The Jones–Wenzl recursion in general.* Write $\mathrm{ptr}(X)$ for the closure of the last strand of X on the right, a map from n strands to $n - 1$, and suppose $[m] \neq 0$ for $m \leq n + 1$. Suppose p_n is an idempotent killed by $U_1, \dots, U_{n-1}$ (on both sides), with $(p_{n-1} \otimes 1)p_n = p_n = p_n(p_{n-1} \otimes 1)$ and $\mathrm{ptr}(p_n) = -\frac{[n+1]}{[n]}\,p_{n-1}$.

(i) Show that $U_n(X \otimes 1)U_n = (\mathrm{ptr}(X) \otimes 1 \otimes 1)\,U_n$ for X on n strands (draw it for $n = 2$), and deduce that $p_{n+1} = p_n \otimes 1 + \frac{[n]}{[n+1]}(p_n \otimes 1)U_n(p_n \otimes 1)$ is killed by U_n.

(ii) Show that p_{n+1} is killed by $U_1, \dots, U_{n-1}$, is idempotent, and is absorbed by $p_n \otimes 1$.

(iii) Show that $\mathrm{ptr}(p_{n+1}) = -\frac{[n+2]}{[n+1]}p_n$, and conclude by induction that the closure of p_n is $(-1)^n[n + 1]$.

Hint. (i) Close the last strand of X through the cup and cap of U_n. (iii) $\mathrm{ptr}(p_n \otimes 1) = \delta p_n$, and $[2][n + 1] = [n + 2] + [n]$.

Answer. (i) $U_np_{n+1} = U_n(p_n \otimes 1) - (p_{n-1} \otimes 1 \otimes 1)U_n(p_n \otimes 1) = 0$ by absorption. (ii) $U_ip_n = 0$ for $i < n$ kills both terms; $p_{n+1}^2 = p_{n+1}(p_n \otimes 1) = p_{n+1}$. (iii) $\mathrm{ptr}(p_{n+1}) = (\delta + \frac{[n]}{[n+1]})p_n = -\frac{[n+2]}{[n+1]}p_n$; closures multiply by $-\frac{[n+2]}{[n+1]}$ from 1.